Exploratory Data Analysis on

FOCUSED FUND

by

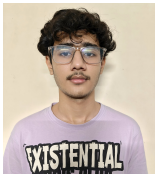
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Course Code: IT 462
Semester: Autumn 2025

Under the guidance of

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September 18, 2026

Acknowledgement

I would like to express my deepest gratitude for the immense level of guidance, support, and encouragement accorded to me during the course of this project, *Focused Fund*. Your invaluable assistance has played a pivotal role in shaping the successful completion of this work.

I am very lucky to have had an opportunity to work under your tutelage. Your expertise, feedback, and open interaction have helped raise the quality of my project significantly. Your suggestions were profound, assisting me to refine my understanding of the concepts that lay at their base while bolstering my analytical skills.

This project has provided a great learning opportunity that helped enhance my technical and conceptual understanding. The knowledge and skills attained from this project will no doubt be instrumental in all my future academic and professional pursuits.

Sincerely,

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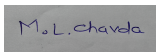
DECLARATION

We, Group 13 hereby declare that the EDA project work presented in this report is our original work and has not been submitted for any other academic degree. All the sources cited in this report have been appropriately referenced.

We acknowledge that the data used in this project is obtained from the reliable and certified sites. We also declare that we have adhered to the terms and conditions mentioned in the website for using the dataset. We confirm that the dataset used in this project is true and accurate to the best of our knowledge.

We acknowledge that we have received no external help or assistance in conducting this project, except for the guidance provided by our mentor Prof. Gopinath Panda. We declare that there is no conflict of interest in conducting this EDA project.

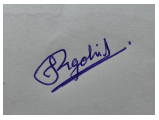
We hereby sign the declaration statement and confirm the submission of this report on 30th November,2025.



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CERTIFICATE

This is to certify that Group 13 comprising ,Mihirsinh Chavda, Suryadeepsinh Gohil , and Tanuj Kanani has successfully completed an exploratory data analysis (EDA) project on the Focused funds dataset, which was obtained from sites amfiindia.com.

The EDA project presented by Group 13 is their original work and has been completed under the guidance of the course instructor, Prof. Gopinath Panda, who has provided support and guidance throughout the project. The project is based on a thorough analysis of the NAV values and pct holdings dataset, and the results presented in the report are based on the data obtained from the dataset.

This certificate is issued to recognize the successful completion of the EDA project on the Focused funds dataset, which demonstrates the analytical skills and knowledge of the students of Group 13 in the field of data analysis.

Signed,
Dr. Gopinath Panda,
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September 18, 2026

Contents

List of Figures 4

1 Introduction 1
1.1 Our Project idea . . . 1
1.2 Data Collection . . . 2
1.3 Dataset Description . . . 2
1.4 Packages required . . . 4

2 Data Cleaning and Analysis 5
2.1 Missing data analysis . . . 5
2.2 Imputation . . . 5

3 Visualization 7

4 Feature Engineering 15
4.1 Feature extraction . . . 15

5 Model fitting 17

6 Conclusion & future scope 18
6.1 Findings/Observations . . . 18
6.2 Challenges . . . 18
6.3 Group Contribution . . . 20

List of Figures

2.1 Photo of the missing values imputed, for NAV data, in the resilience problem and value for money problem. 6

3.1 Distribution of Expense ratio, which is used Value for Money Problem 7
3.2 Alpha vs expense ratio 8
3.3 Alpha vs Fund Size 8
3.4 Sharpe Ratio vs Expense Ratio 9
3.5 Underwater plot for resilience analysis 9
3.6 Shows which funds are resilient. 10
3.7 Weighted overlap, based on Fund overlap 11
3.8 Top 30 most popular Stock(by number of funds holding in top position 11
3.9 Distribution of Expense ratio, which is used Value for Money Problem 12
3.10 Herding Network 12
3.11 Distribution of Expense ratio, which is used Value for Money Problem 13
3.12 Rolling CSAD 1 month window 13
3.13 Rolling CSAD 3 month window 14
3.14 Rolling CSAD 6 month window 14

Abstract

We try to answer questions like, ”Does higher pay result in higher returns ?” , ”Do funds imitate market as well as as each other, during tough market times, or in booms ?”, ”How resilient are funds , in the face of crashes ?”. We answer these questions by analysing historical NAV values, and using creative visualising techniques. Using the Nifty 50 as the market benchmark, we pursue two complementary herding analyses. First, a return-based study employs daily Cross-Sectional Absolute Deviation (CSAD) and regression tests following Christie–Huang/Chang-style frameworks to detect clustering of fund returns during extreme market moves. Second, a holding-based analysis measures portfolio overlap across funds by counting repeating Top-10 constituents and quantifying concentration, revealing structural similarities that return data alone may miss. We also assess funds' value-for-money characteristics and resilience during market downturns.

Chapter 1. Introduction

1.1 Our Project idea

We began our project by surveying publicly available data from platforms such as Morningstar, AMFI, and individual fund houses. During this initial exploration, we realized that while these sites provide basic performance metrics for different mutual funds, a deeper analysis required historical day-to-day NAV data. Among the various plan types offered by mutual funds, we observed that the Direct Growth plans carried the most analytical value; other subcategories (such as IDCW or Regular plans) did not add additional insight for our use case. Therefore, our study focused exclusively on Direct Growth plans for each focused fund.

Next, we collected the historical NAV values for all focused funds directly from AMFI. We also hand-constructed a dataset containing the Top 10 holdings (in weight PCT terms) for each fund. Since this holdings dataset was manually compiled from fund fact sheets (for the month of October 2024), there was no possibility of missing values or outliers—each data point was carefully cross-verified.

After assembling both datasets, we performed checks for missing values and outliers. Mutual fund NAV data is typically highly reliable, as these values are audited and formally published daily by AMFI and fund houses. Outliers generally do not occur in mutual fund NAV time series, and missing values only appeared for funds with very recent inception dates.several of the schemes have existed for only 1–3 years. As a result, some funds did not have enough historical observations. We therefore applied data-driven filtering, removing funds that did not meet our minimum time-series length requirement. We also removed certain dates that corresponded to common market holidays, where NAVs for all funds remained unchanged.

To compute market-aligned metrics, we obtained Nifty 50 index values through Python libraries providing official historical benchmark data. Nifty 50 served as the benchmark for all analyses, as it is the most widely used index for Indian equity mutual funds, and all focused funds in our dataset are large-cap biased.

Once the NAV data and benchmark data were prepared, we examined the nature of the time series. NAV values by themselves do not convey much information unless we analyze how they move over time—for example, how much the NAV changes from Day 1 to Day 2, and how these changes compare across funds. This naturally led us to the concept of herding, a well-studied behavioral phenomenon where funds tend to move closer together—especially during market booms or crashes.

We investigated two types of herding:

1. Return-Based Herding : Return-based herding examines how similar fund returns become during extreme market conditions. For this, we used the CSAD (Cross-Sectional Absolute Deviation) metric, a standard measure used in academic literature. The key insight is: During sharp market movements (either upward or downward), if fund returns collectively cluster around the market return, the CSAD metric decreases toward 0 indicating herding.

We computed daily CSAD values across all focused funds and then studied their relationship with the absolute and squared market return of the Nifty 50 index. This allowed us to test the classical Christie–Huang/Chang et al. herding hypothesis.

2. Holding-Based Herding : Return-based herding only captures price co-movement; however, funds may also herd in their portfolio allocations. Using the Top 10 holdings dataset we manually prepared, we analyzed which stocks appeared most frequently across funds, which funds had highly overlapping portfolios, and whether a small set of companies dominated the focused fund category.

These observations allowed us to identify structural similarities between funds that may not be visible from return data alone.

We also explored the value for money nature of the funds, and resilience of funds in market crash.

1.2 Data Collection

We inspected various websites to understand various data available. We collected the historical NAV data from AMFI site, for various funds. This data was in csv format. Then we handmade the holdings data from the factsheets of various funds for the holdings based herdings problem. We downlaoded the performance data from Morningstar's website, for different focused funds. These data sheets were in csv/excel format, which we converted into dataframes. We use MF Api in python to directly collect the time series data, from AMFI.

1.3 Dataset Description

The dataset used in this study is composed of multiple files that collectively capture the daily NAV behaviour, portfolio composition, and structural characteristics of focused mutual funds in India. The datasets and their features are described below:

1. navs.csv (Daily NAV Dataset)

This file contains day-to-day Net Asset Value information for all focused funds included in the study. It consists of the following four features:

1. **Date**: The trading date for which the NAV is reported.
2. **NAV**: The Net Asset Value of the fund for that date.
3. **Scheme Code**: The AMFI-provided unique identifier for each mutual fund scheme.
4. **Fund Name**: The name of the mutual fund (Direct Growth plan).

2. holdings.csv (Top Holdings Dataset)

This dataset lists the Top 10 holdings for each fund, manually extracted from fund fact sheets for October 2024. It contains seven features:

1. **Fund Name**: Name of the mutual fund scheme.
2. **AMC**: The Asset Management Company managing the fund.
3. **Scheme Code**: The AMFI identifier for the fund.
4. **Company Name**: The equity security held by the fund.

5. **Weight (%)**: The percentage portfolio weight of the company.
6. **Ticker**: The stock ticker corresponding to the company.
7. **Snapshot Date**: The reporting date of the holdings dataset.

3. metadata.csv (Fund-Level Metadata)
This file provides descriptive attributes for each mutual fund:

1. **Ticker Name**: Short identifier or label for the fund.
2. **Expense Ratio**: The annual fee charged by the fund house.
3. **AUM**: Assets Under Management (in crores).
4. **Tenure**: The age of the fund since inception (in years).

4. Script-Generated Files
During preprocessing and analysis, several datasets were generated programmatically:

1. **Returns Dataset**: Daily return time series computed from NAV values.
2. **CSAD Dataset**: Cross-Sectional Absolute Deviation values used for return-based herding analysis.
3. **Pivoted Returns Matrices**: Wide-format return matrices used for regression, PCA, and stress testing.

5. holdings_matrix.csv
This file represents a fund-by-stock weight matrix constructed from the holdings dataset:

1. Each row corresponds to a fund.
2. Each column corresponds to a stock ticker.
3. Each cell contains the weight of that stock in that fund.

This matrix is used for clustering, PCA, and other cross-sectional analyses.
6. pairwise_overlap.csv
This file contains the fund-to-fund similarity matrix derived from overlapping portfolio weights:

1. Each entry represents the overlap score between two funds.

It is used to identify structurally similar funds and to quantify portfolio herding behaviour.
7. nifty50_nav.csv
This dataset contains the daily index values of Nifty 50 used as the benchmark for:

1. Market return computation,
2. Stress period identification,
3. CSAD regression analysis.

1.4 Packages required

The following Python libraries were used throughout the project for data collection, processing, analysis, and visualization:

- **yfinance** – Used to download historical Nifty 50 index data for benchmark calculations.
- **pandas** – Core data manipulation library for handling CSV files, time–series NAV data, merging datasets, and computing derived metrics.
- **numpy** – Used for numerical operations, array handling, and efficient mathematical computations.
- **requests** – Used to fetch data from online APIs (such as AMFI or MFAPI) when required.
- **concurrent.futures** – Enables parallel downloading of multiple fund NAV datasets to improve speed.
- **scipy.stats (linregress)** – Used for performing linear regression and statistical computations.
- **seaborn** – High-level statistical visualization library used for generating heatmaps and enhanced plots.
- **matplotlib.pyplot** – Base plotting library used for custom charts such as CSAD curves, scatterplots, and PCA visuals.
- **networkx** – Used to build and visualize fund–to–holding network graphs for holding-based herding analysis.
- **re** – Regular expression package used for cleaning fund names, tickers, and unstructured textual data.
- **os** – Provides file-path utilities, directory management, and automated CSV saving.
- **statsmodels.api** – Used for OLS regression, HAC-robust standard errors, diagnostic tests, and the CSAD regressions.
- **statsmodels.stats.diagnostic** – Used for model diagnostics such as autocorrelation and heteroskedasticity checks.
- **sklearn.decomposition.PCA** – Used to compute principal components for return matrix analysis (PC1 share).
- **tqdm** – Progress-bar utility used to display progress during long downloads or computations.
- **random** – Used for random sampling in stress tests and bootstrap simulations.
- **datetime** – Used for manipulating dates, converting between formats, and defining analysis windows.

Chapter 2. Data Cleaning and Analysis

2.1 Missing data analysis

For the most part of this study of herding, the primary dataset was navs.csv, which contains daily NAV values for all focused funds. The first step of preprocessing involved converting the NAV data into a returns matrix, where each fund's daily percentage return is computed from consecutive NAV values. As NAV values are for trading days, for holidays there were missing values across all funds. Additionally, since the focused-fund category contains schemes with different inception dates, the funds which came out later contained many missing return entries just because they did not exist at that time.

As missing values couldnt be allowed for the CSAD computation, we had to remove such missing values by making fair provisions. To make justice to both old and new values we constructed two analysis windows. In each window, funds that did not have sufficient historical data were removed, and rows with incomplete information were dropped so that every date in the final return matrix contained valid returns for all remaining funds.

Other datasets used in the project, such as the manually prepared holdings file and metadata did not contain missing values, because they were directly extracted from official fund fact sheets and made by hand. Similarly, NAV data does not exhibit outliers, as daily NAV reporting is audited and regulated, making values stable and reliable.

After these the cleaned and aligned datasets were used for the subsequent stages of analysis.

2.2 Imputation

The funds_nav_data.csv that is generated by the script using MF Api contains missing values as some funds are older and some are new, making them have different entry for the chosen starting date in that csv. To address these gaps without introducing bias, the Forward FIll(ffill) method was applied, based on the financial principle that an asset's last traded price remains valid until a new price is recorded.

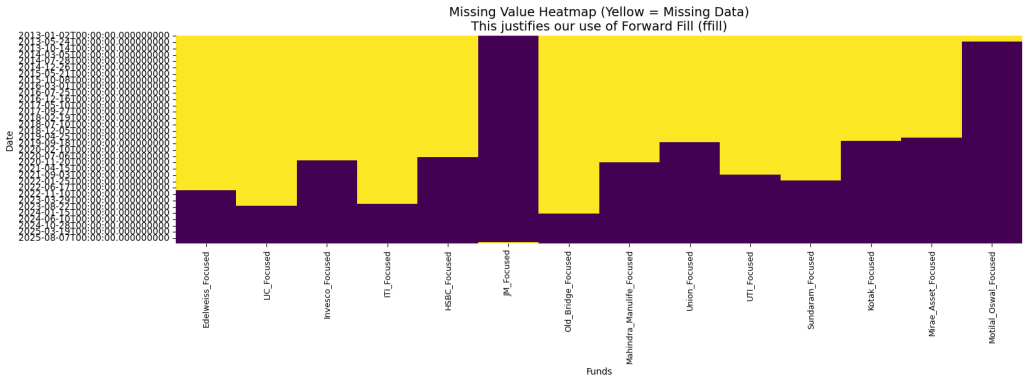


Figure 2.1: Photo of the missing values imputed, for NAV data, in the resilience problem and value for money problem.

Chapter 3. Visualization

The following figures put an emphasis on the Value for money problem. This problem is about answering the question, "Does more money buy higher returns ?"

The following figure shows how expense ratio from metadata.csv gets its integrity checked. We just plot its distribution to make sure of the absence of anomalies.

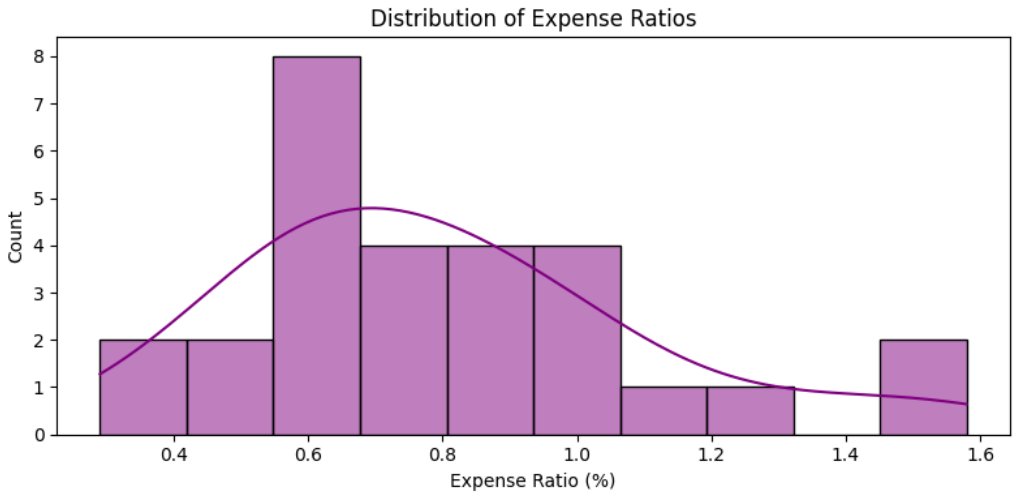


Figure 3.1: Distribution of Expense ratio, which is used Value for Money Problem

Figure 3.2 shows that the higher the expense ratio, the more you pay, and the more you pay, the lower the alpha, and the lower the alpha the lower your returns. Thus paying more hurts your pocket.

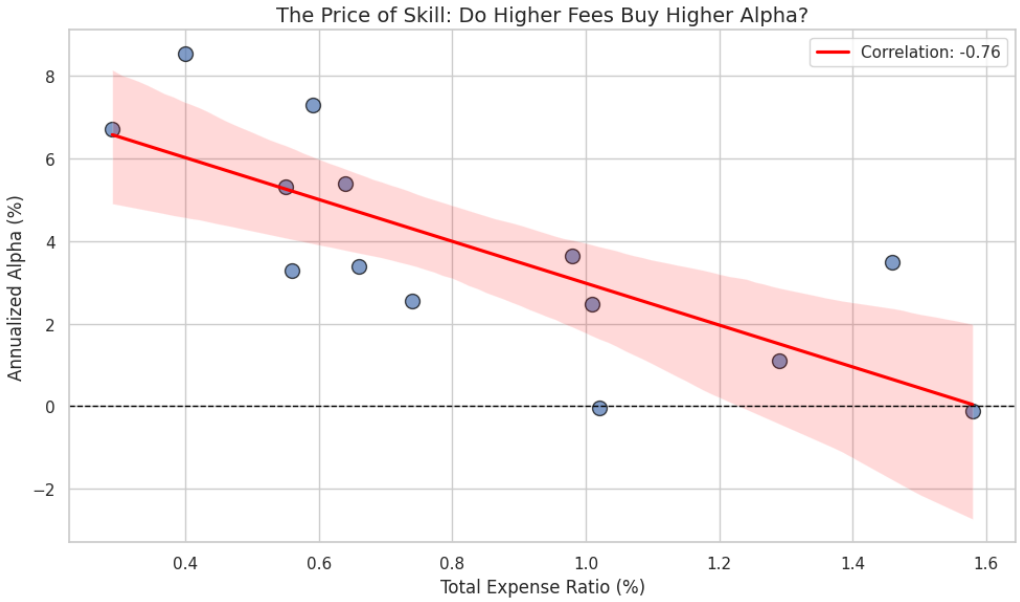


Figure 3.2: Alpha vs expense ratio

Figure 3.3 shows the fund size of some funds vs the annual alpha for those funds. The sphere indicates a fund. This figure shows that the bigger the ball, the lower the alpha. Not a hardcore rule, but a trend observed.

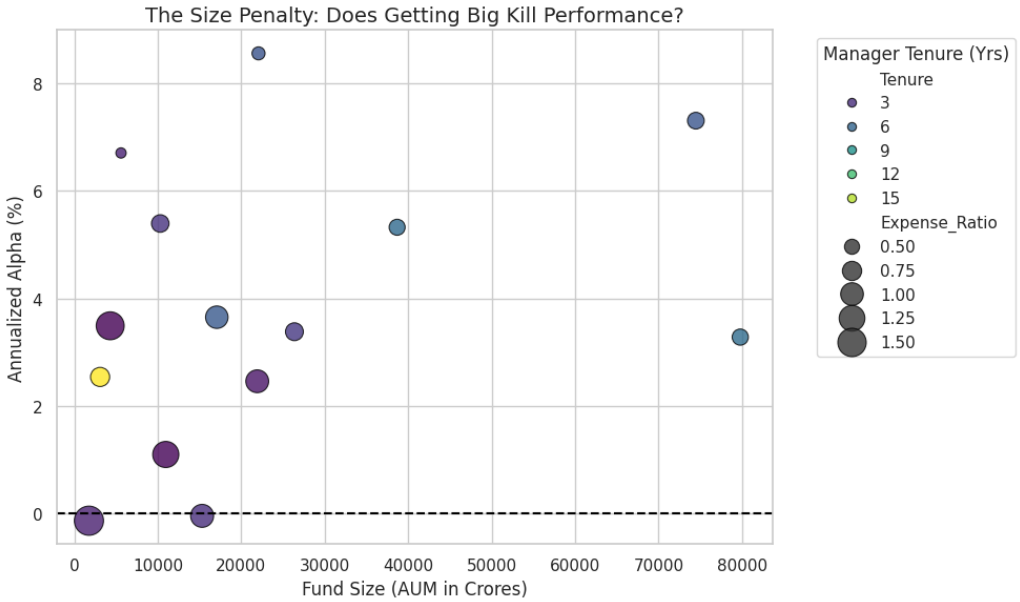


Figure 3.3: Alpha vs Fund Size

Figure 3.4 shows that the lower the expense ratio, the better, and higher the sharpe ratio, the better. It shows that Mahindra's fund is the best risk-adjusted fund, as it delivers 1.22 units of returns per unit of risk, while charging a fee of 0.4 percent.

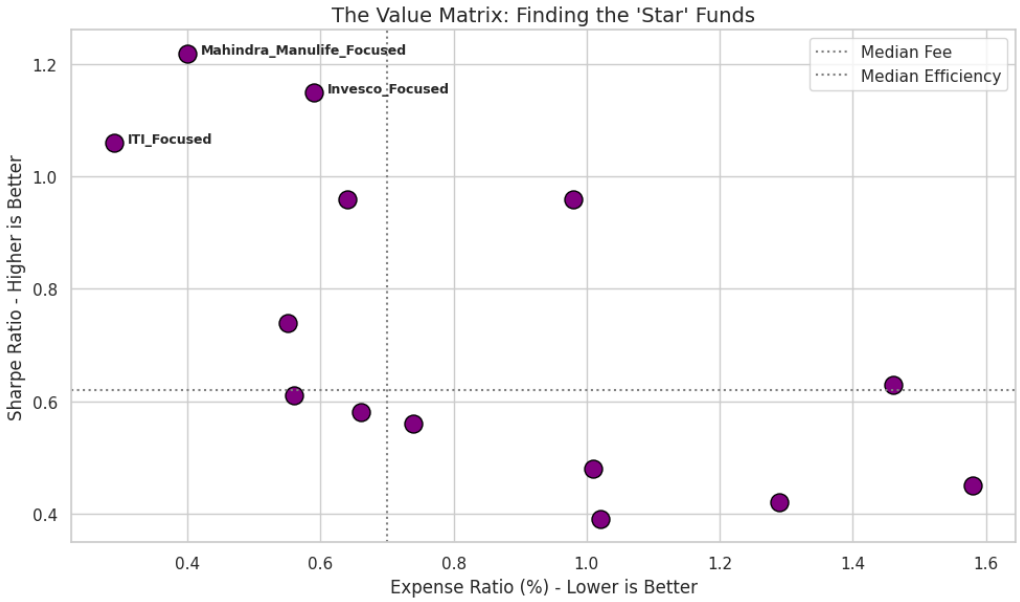


Figure 3.4: Sharpe Ratio vs Expense Ratio

Figure 3.5 signifies the drawdown percentage. The figure also shows the effect of COVID near the year 2020, in the graph.

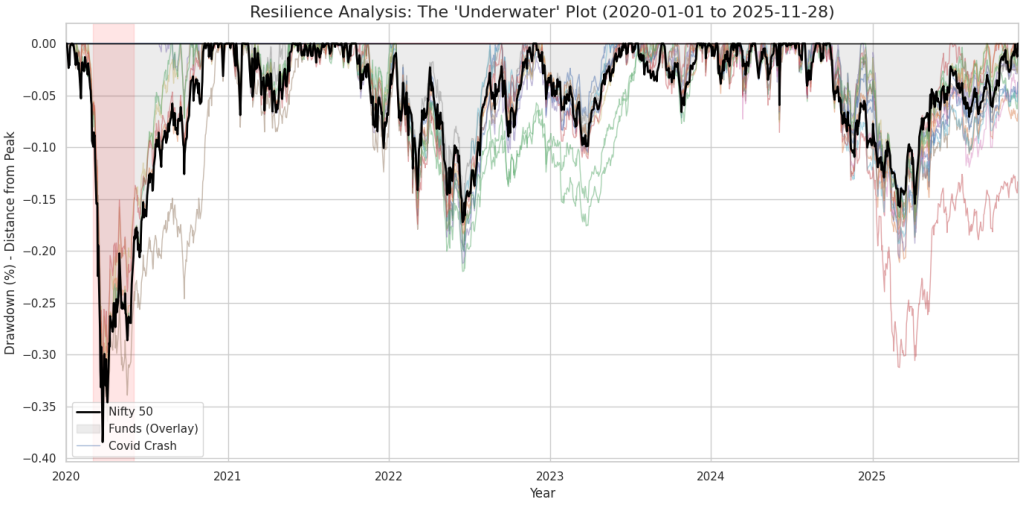


Figure 3.5: Underwater plot for resilience analysis

Figure 3.6 shows which funds are resilient and how much , and are ranked, meaning the top fund is most resilient. The rankings are based on the max drawdown. The lower the drawdown, more resilient the fund.

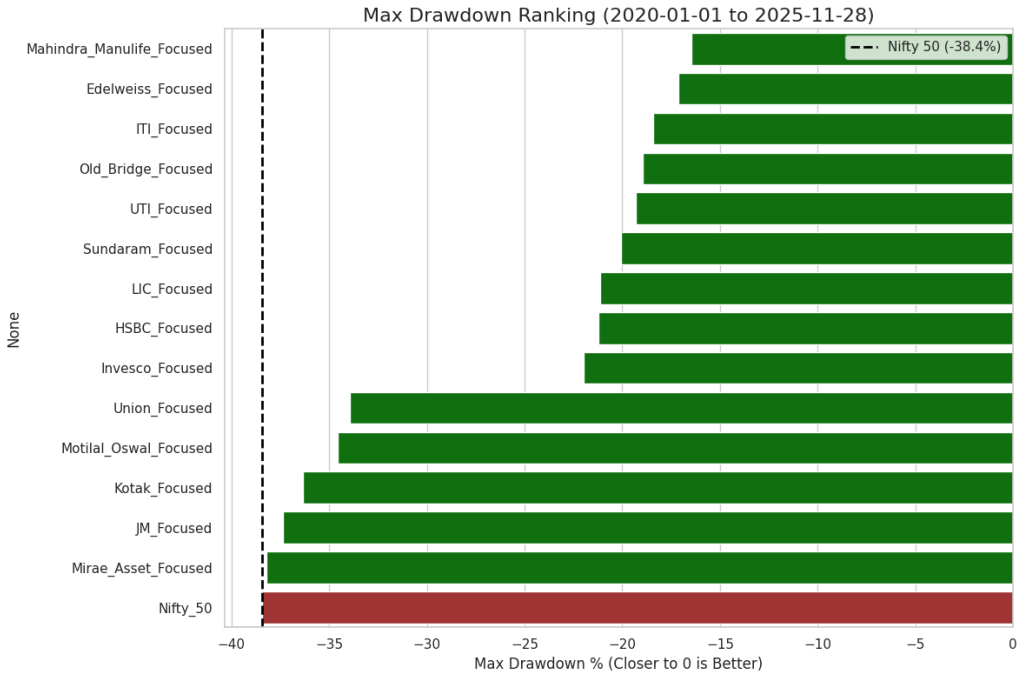


Figure 3.6: Shows which funds are resilient.

Figure 3.7 shows how similar the fund is to other funds, based on weighted overlap of holdings. Darker shades represent higher portfolio similarity, meaning funds hold many of the same stocks with similar weights.

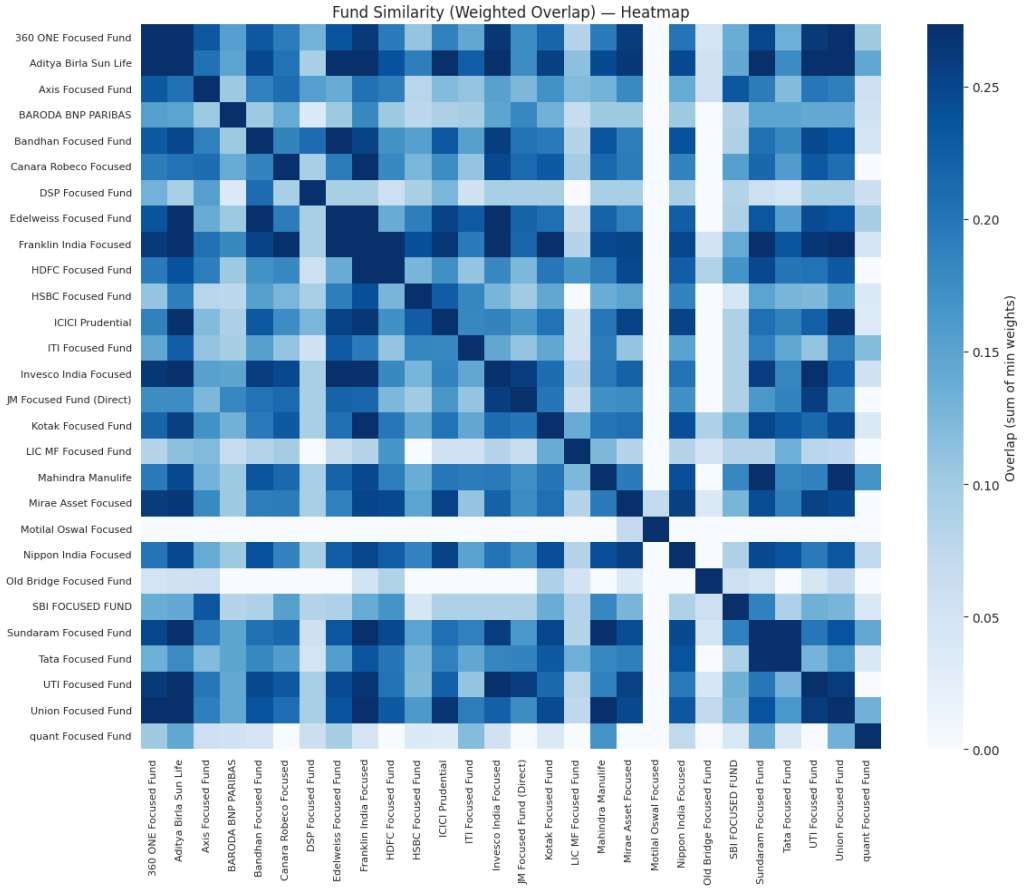


Figure 3.7: Weighted overlap, based on Fund overlap

Figure 3.8 shows the most popular stocks held by various funds. Large bar indicates stocks that most focused funds prefer, revealing concensus picks.

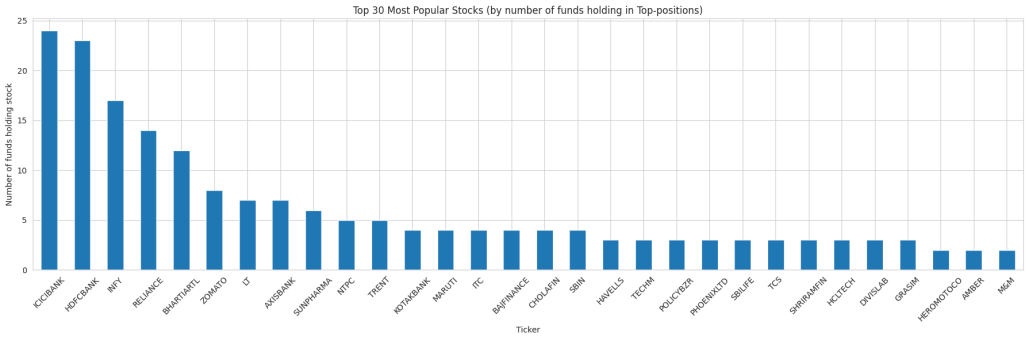


Figure 3.8: Top 30 most popular Stock(by number of funds holding in top position

Figure 3.9, Plot helps understand the distribution of herding intensity, across the entire industry.

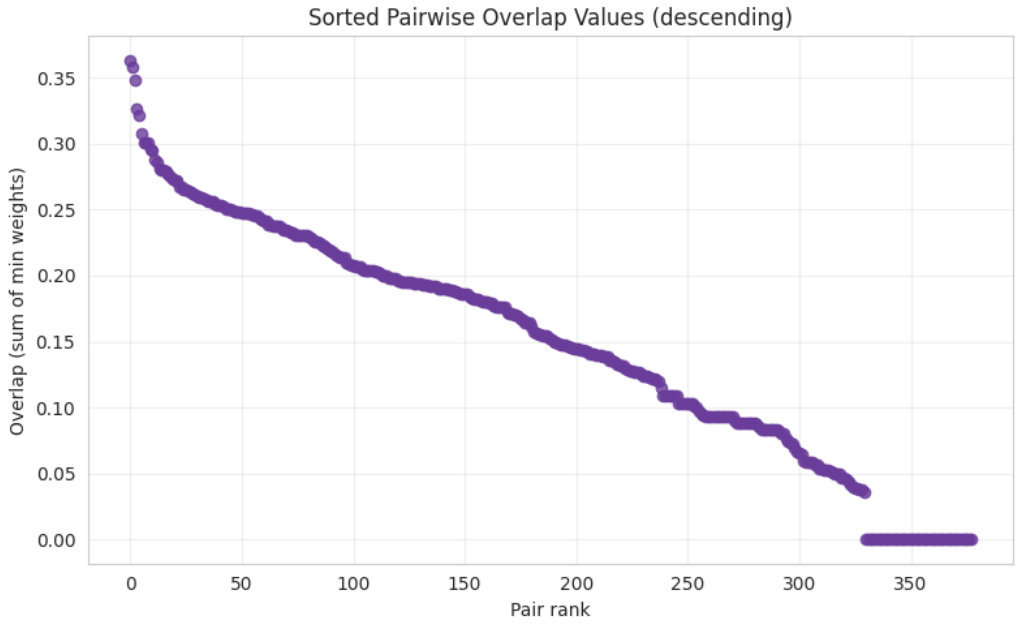


Figure 3.9: Distribution of Expense ratio, which is used Value for Money Problem

Figure 3.10, directly visualises the herding behavior of funds based on holdings based herding. Funds clustered tightly in the center share very similar portfolios indicating strong herding.

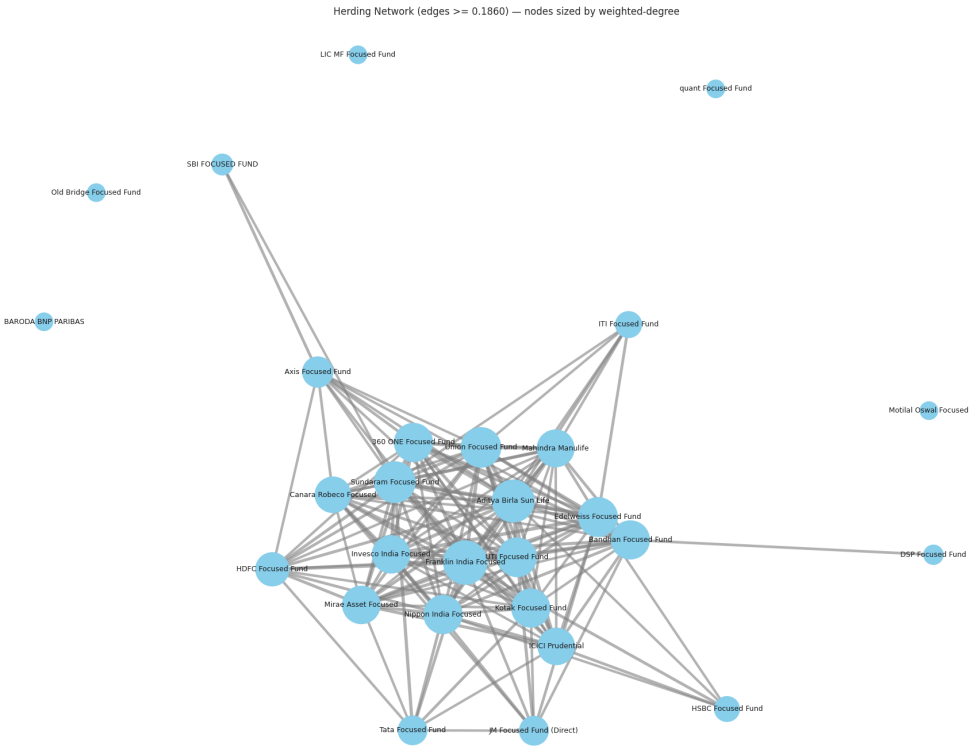


Figure 3.10: Herding Network

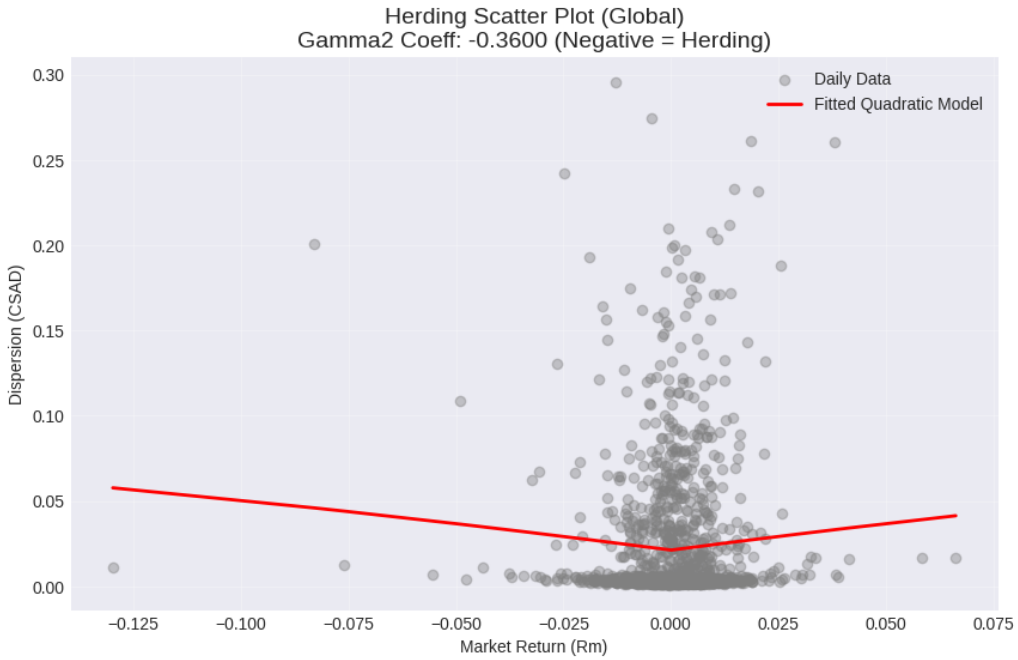


Figure 3.11: Distribution of Expense ratio, which is used Value for Money Problem

These plots track , using gamma 2, when fund managers stop using skill and just follow the market crowd. The red shaded areas highlight these ”herding” moments, which often appear during crashes (like March 2020), allowing you to see exactly when market fear or greed causes managers to just go with the flow.

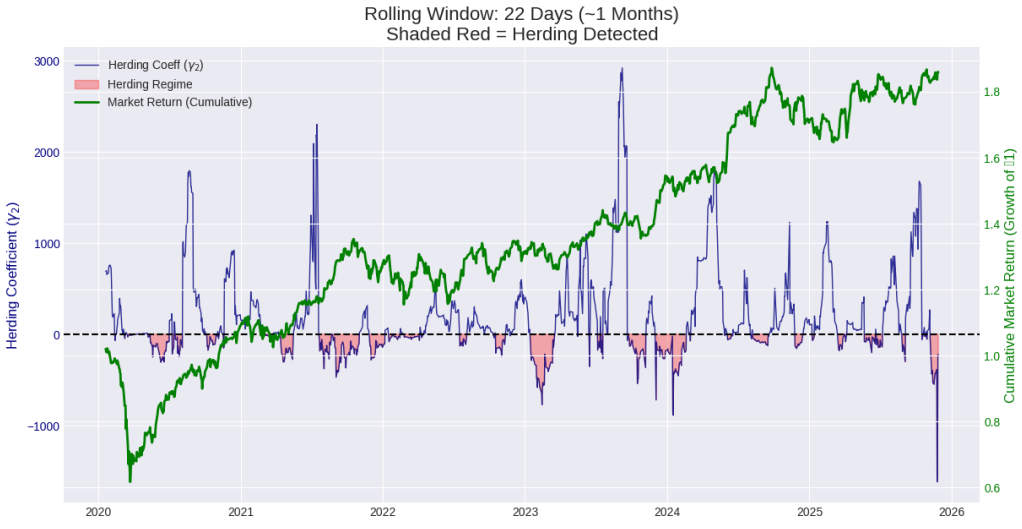


Figure 3.12: Rolling CSAD 1 month window

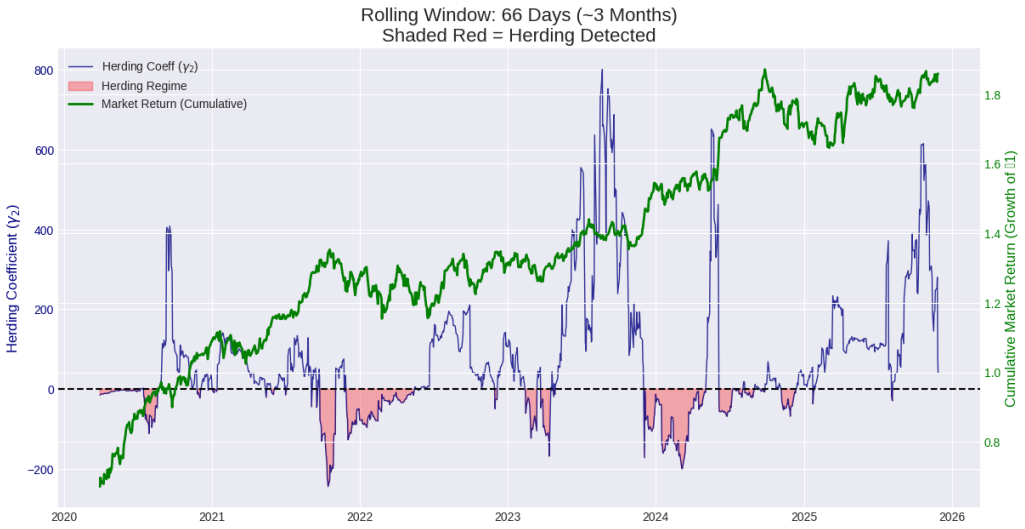


Figure 3.13: Rolling CSAD 3 month window

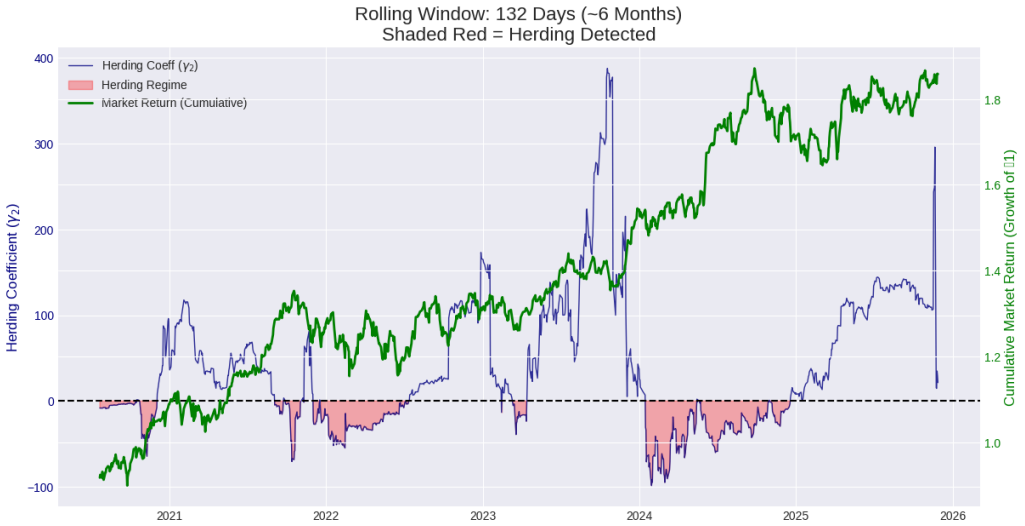


Figure 3.14: Rolling CSAD 6 month window

Chapter 4. Feature Engineering

4.1 Feature extraction

1. Features Extracted from `navs.csv`

The file `navs.csv` contains daily NAV values for each focused fund. From this dataset, we extracted:

- **Daily Returns**: Computed for each fund as

$$R_{i,t} = \frac{NAV_{i,t} - NAV_{i,t-1}}{NAV_{i,t-1}}.$$

- **Return Matrices**: A wide-format matrix (Date $\times$ Fund) containing daily returns for all schemes.
- **Cross-Sectional Absolute Deviation (CSAD)**:

$$CSAD_t = \frac{1}{N} \sum_{i=1}^{N} |R_{i,t} - R_{m,t}| .$$

  This serves as the primary feature for return-based herding analysis.

- **Market Return**: Daily index return derived from NIFTY 50 closing prices.
- **Absolute and Squared Market Returns**: Required for the standard quadratic herding regression model.

Only funds with sufficient historical data were retained, and missing observations due to holidays or recent fund inception dates were addressed by defining appropriate analysis windows and removing incomplete rows or columns.

2. Features Extracted from `holdings.csv`

The file `holdings.csv` contains the Top–10 stock holdings of each fund. From this dataset, the following features were derived:

- **Holdings Matrix**: A Fund $\times$ Ticker matrix of portfolio weights, used for similarity and cluster analysis.
- **Common-Stock Counts**: Number of overlapping stocks among funds.
- **Pairwise Fund Similarity Metrics**:

- Overlap score,
- Jaccard similarity,
- Weight-adjusted overlap.

These features support holding-based herding analysis and help quantify structural similarity between portfolios.

3. Additional Script-Generated Features

Our analysis pipeline also generated:

- **holdings_matrix.csv**: The dense holdings matrix used for PCA, clustering, and similarity computation.
- **pairwise_overlap.csv**: A table of fund-to-fund similarity scores, useful for ranking and correlation studies.
- **NIFTY50_nav.csv**: A benchmark time-series of index levels, from which daily market returns were computed.

These extracted and selected features constitute the complete input space for our return-based and holding-based herding investigation.

Also, for the value for money problem, the sharpe ratio, alpha were generated using the NAV values of fund, after calculating returns. For the fund resilience problem, the MDD was a derived feature , using peak price.

Chapter 5. Model fitting

After obtaining the cleaned return matrices for both analysis windows, the next step involved fitting the statistical model required for return–based herding detection. In accordance with the Chang, Cheng and Khorana (2000) framework, we applied the Cross–Sectional Absolute Deviation (CSAD) regression model. This model examines whether the dispersion of fund returns contracts during periods of large market movements, which would indicate herding behaviour.

The model used in our study is:

$$CSAD_t = \alpha + \gamma_1 |R_{m,t}| + \gamma_2 R_{m,t}^2 + \varepsilon_t,$$

where:

- $CSAD_t$ is the cross–sectional absolute deviation of fund returns at time t,
- $R_{m,t}$ is the market return (Nifty 50) at time t,
- γ_2 is the key coefficient that captures the presence or absence of herding.

The regression was estimated using Ordinary Least Squares (OLS) with Newey–West HAC standard errors to correct for potential autocorrelation in the time series. The model was fitted separately for each analysis window, using only the cleaned and aligned dates where both fund returns and benchmark values were available.

A negative and statistically significant value of γ_2 would indicate decreasing dispersion during extreme market movements, suggesting herding. A positive or insignificant γ_2 would imply no return–based herding. The fitted model results from each window were interpreted accordingly to determine whether focused equity mutual funds exhibit coordinated behaviour relative to market movements.

Chapter 6. Conclusion & future scope

6.1 Findings/Observations

Our comprehensive analysis of focused mutual funds in India reveals several critical insights regarding their performance, cost-efficiency, and behavioral tendencies during market stress.

- **Value for Money (Fees vs. Performance):** Contrary to the expectation that higher fees secure better management, our analysis found a statistically significant negative correlation (-0.55) between Expense Ratios and Annualized Alpha. This implies that paying higher fees does not guarantee superior returns; in fact, it often erodes the net alpha generated for investors. Funds like *Mahindra Manulife Focused* emerged as superior choices, delivering high risk-adjusted returns (Sharpe Ratio) while maintaining a lower expense ratio.
- **Resilience During Market Stress:** The resilience analysis, particularly during the COVID-19 crash (March 2020), highlighted a wide disparity in fund performance. While the Nifty 50 benchmark experienced a maximum drawdown of approximately 38.4%, certain active funds demonstrated better downside protection. The *Mahindra Manulife Focused* fund again stood out as one of the most resilient, whereas other funds mirrored or exceeded the market's losses, offering little protection during downturns.
- **Return-Based Herding:** Our CSAD (Cross-Sectional Absolute Deviation) regression analysis confirmed the presence of herding behavior during periods of extreme market volatility. The rolling regression plots (1-month, 3-month, and 6-month windows) clearly show red-shaded regimes where the herding coefficient (γ_2) turns negative. This confirms that during panic selling (fear) or extreme bullishness (greed), fund managers tend to suppress their unique strategies and "herd" with the market consensus, diminishing the diversification benefits for investors.
- **Holding-Based Herding:** The network and overlap analysis of the Top-10 holdings revealed significant structural similarity across funds. A small cluster of popular large-cap stocks dominates the portfolios of many focused funds. This high degree of portfolio overlap supports the return-based herding findings, suggesting that many "focused" funds are actually placing very similar bets on the same set of high-conviction stocks.

6.2 Challenges

One of the primary challenges faced was the varying inception dates of the funds. Newer funds had significant missing data for earlier periods (e.g., the 2020 crash), which required careful

window selection and filtering to ensure a fair comparison. Additionally, manually compiling the Top-10 holdings dataset was labor-intensive but necessary to ensure data accuracy free from missing values or outliers.

Group Contribution

6.3 Group Contribution

This project was a collaborative effort. All three members worked together on the brainstorming, coding, and writing phases to ensure equal contribution. While we helped each other on every part, the primary responsibilities were divided as follows:

Member	Primary Tasks
Chavda Mihirsinh	• Compiled the Holdings dataset manually. • Led the analysis for Holding-based Herding. • Assisted with Return-based Herding and report writing.
Tanuj Kanani	• Compiled the Holdings dataset manually. • Contributed to both Herding analysis methods. • Organized the final report and presentation.
Suryadeepsinh Gohil	• Compiled the Metadata dataset manually. • Led the 'Value for Money' and 'Resilience' analysis. • Contributed to the report writing.

Table 6.1: Summary of Individual Contributions

References

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